

Motivation: Is Attention All A Time Series Needs?

Before 6-4, **NO!**

With Time Series, Transformers **struggle** due to:

High **computational cost** of $O(N^2)$

Weak local **semantic modeling** (point-wise tokens)

Patching

Segment time series points into **subseries-level tokens**

Benefits

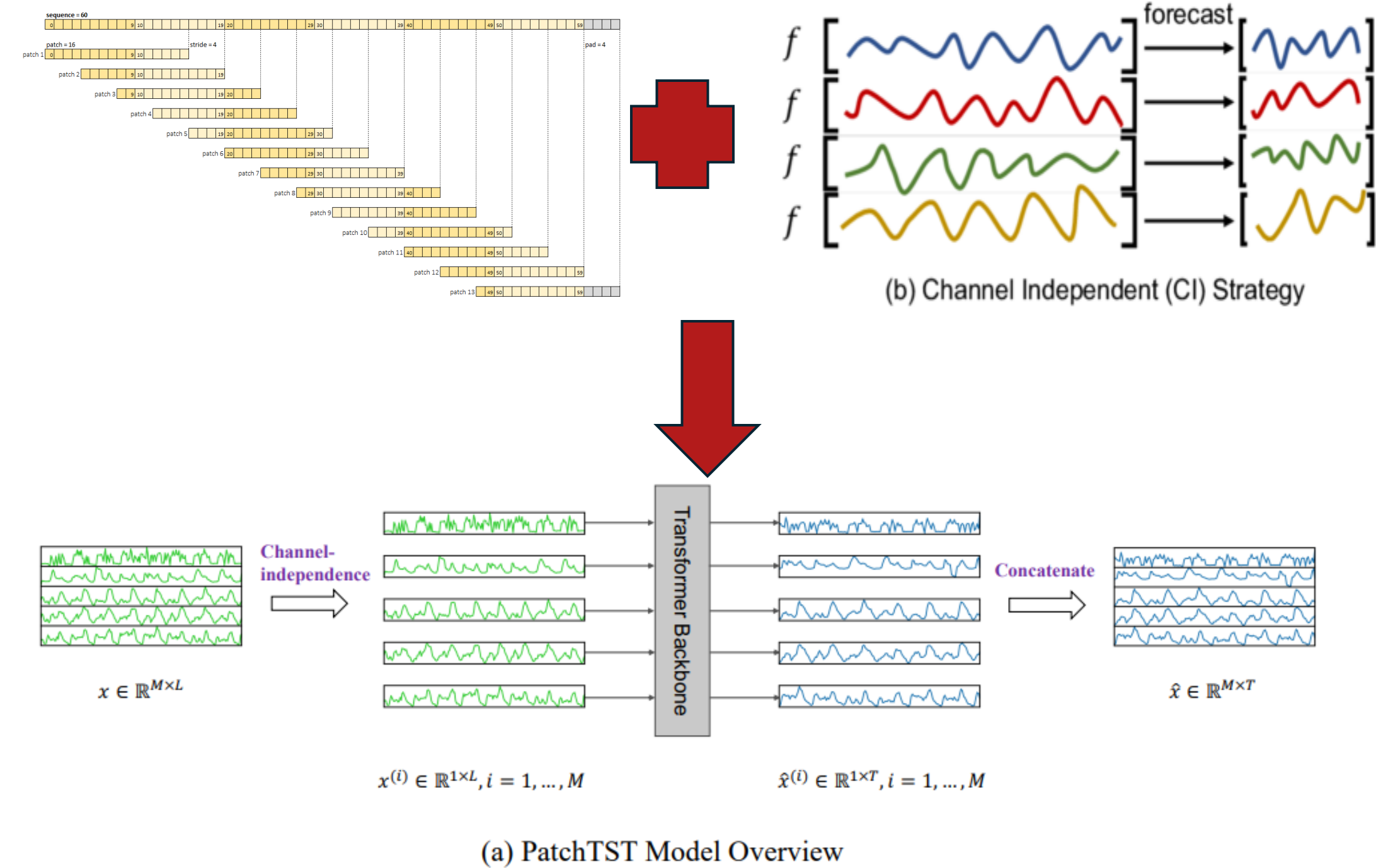
Captures **local temporal patterns**
Shorter **sequence length**
Longer look-back windows

Channel-Independence

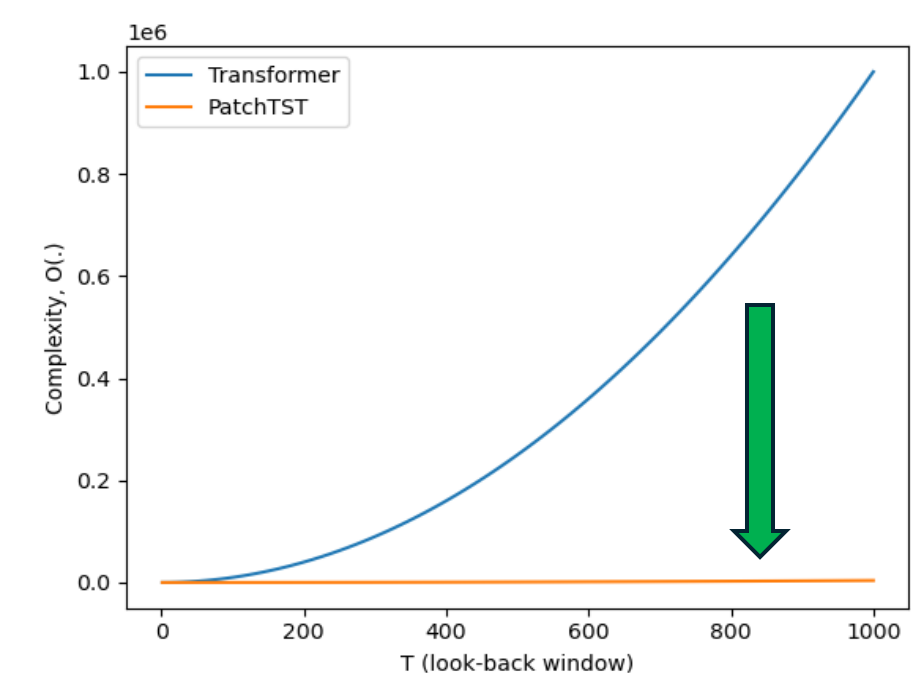
Treat each variate as a separate univariate series **sharing the same weights**

Benefits

Reduces **overfitting**
More **robust to noisy channels** — noise stays isolated.
Scales to **higher-dimensional data**



Patch to Unlock Efficiency



Attention complexity reduces from $O(L^2)$ to $O((\frac{L}{S})^2)$ — quadratic speedup.

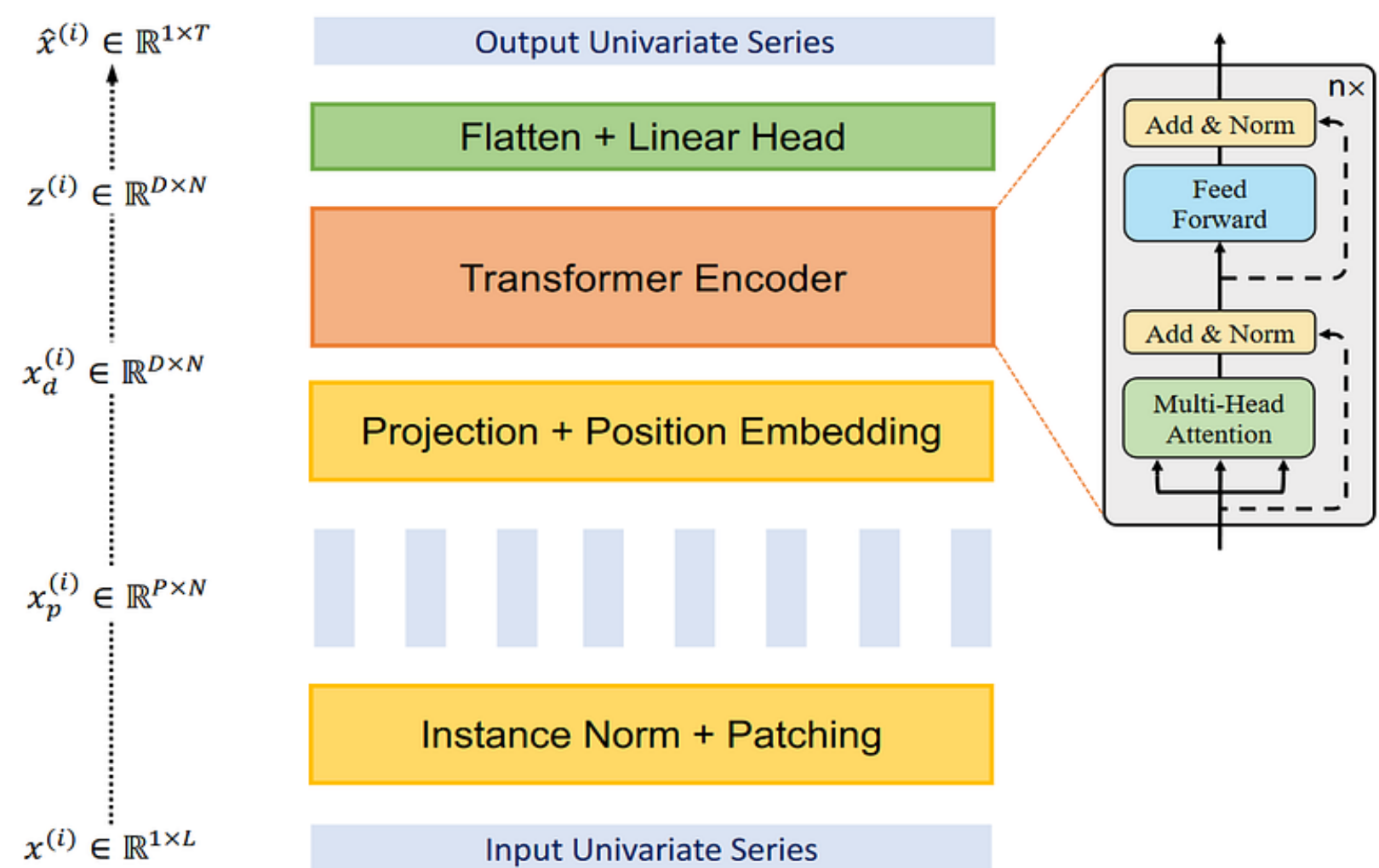
Traffic $\times 22$

Electricity $\times 19$

Weather $\times 4$

Training speedup with patching vs. no patching ($L = 336$)

Model Architecture



Note: Each channel is processed independently then concatenated.

PatchTST splits multivariate input into M independent channels, segments each into patches (length P , stride S), and feeds them into a shared Transformer encoder. This reduces tokens from L to $\approx \frac{L}{S}$. A linear head produces the T-step forecast per channel.

When In Doubt Simplify!

DLinear – A simple linear model that decomposes time series into trend and seasonal components

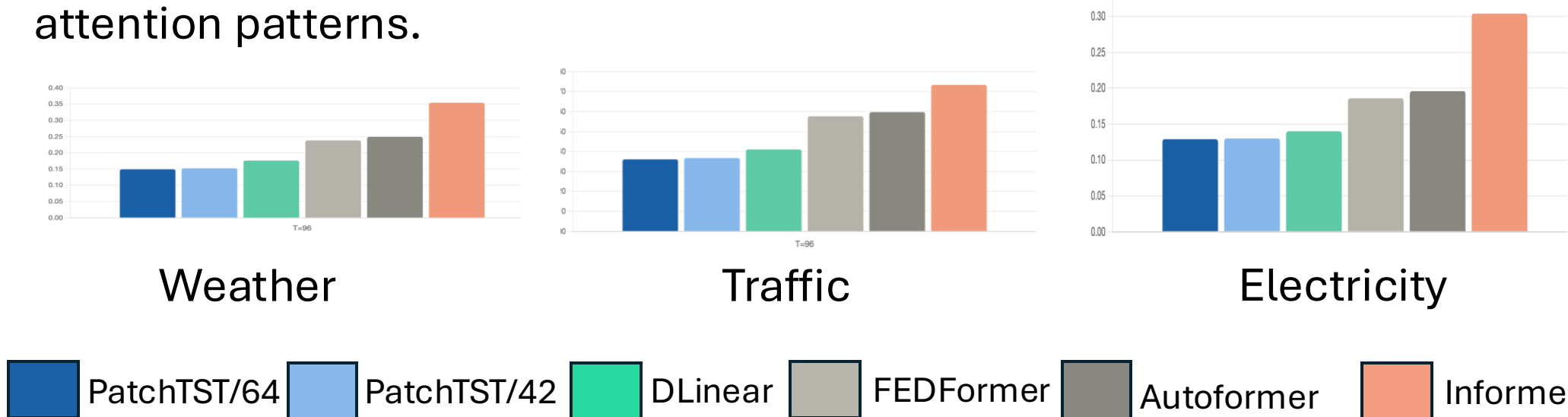
FEDFormer – A Transformer variant that uses frequency-domain decomposition and Fourier-enhanced attention.

Autoformer – A Transformer model that incorporates series decomposition and auto-correlation mechanisms.

Informer – A Transformer designed for long sequences that uses ProbSparse.

Pyraformer – A hierarchical Transformer that models time series at multiple temporal resolutions using pyramidal attention.

LogTrans (LongTrans) – A Transformer that employs logarithmic sparse attention patterns.



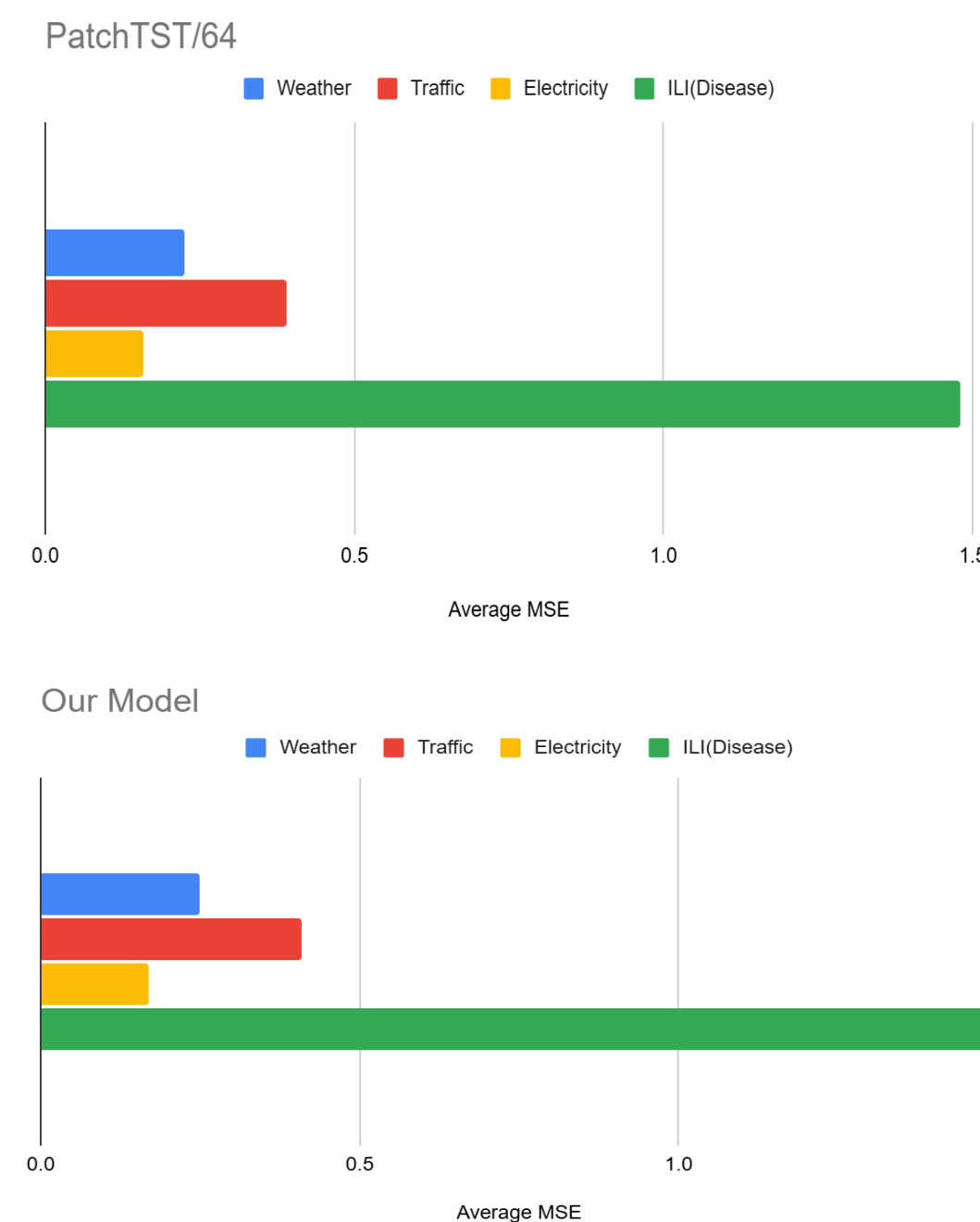
Best MSE — Weather T=96
0.149
vs 0.176 DLinear (+18% better)

Best MSE — Traffic T=96
0.360
vs 0.733 Informer (2x better)

#1 rank across all settings
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PatchTST/64 wins every horizon

From the Paper: Multivariate long-term forecasting results with supervised PatchTST. We use prediction lengths $T \in \{24, 36, 48, 60\}$ for ILI dataset and $T \in \{96, 192, 336, 720\}$ for the others.

Did We Beat IBM Research's Model?



A comparison of the average mean squared error of our trained model vs the official PatchTST/64 model on the four different datasets. Our model performed very similar only being slightly worse on ILI dataset.

A Taste of 6-4

Our Model	Weather		Traffic		Electricity	
	MSE	MAE	MSE	MAE	MSE	MAE
96	0.166	0.211	0.387	0.269	0.140	0.233
192	0.216	0.255	0.399	0.274	0.151	0.242
336	0.275	0.298	0.406	0.278	0.171	0.273
720	0.344	0.346	0.444	0.296	0.209	0.298

Our Model	ILI(Disease)	
	MSE	MAE
24	1.952	0.942
36	1.782	0.887
48	2.016	0.963
60	1.948	0.937

Our Model	FOMC	
	MSE	MAE
12	0.645	0.445
24	0.881	0.625
36	1.017	0.688

So What?

PatchTST: Simple architectural changes outperform complex attention mechanisms.

Better than even the best non-transformer models, **PatchTST** sets a foundation for **Efficient long-horizon forecasting** and **Transferable temporal representations**

References

Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). A time series is worth 64 words: Long-term forecasting with transformers. *International Conference on Learning Representations (ICLR 2023)*. arXiv:2211.14730.